

Course Code: CSE 3015	Course Title: ADVANCED NATURAL LANGUAGE PROCESSING Type of Course: Integrated	L- T-P- C	2 -0	2	3
Version No.	1.0				
Course Pre-requisites	CSE 3014 – Fundamentals of Natural Language Processing				
Anti-requisites					
Course Description	This course is an advanced course for Natural Language Processing. As a part of the course, students will be introduced to solving multiple problems in natural language processing, such as sentiment analysis, machine translation, cognitive natural language processing, etc. Topics include: Machine translation, Text summarization, Sentiment analysis, Cognitive NLP, Gaze behaviour, Evaluation Metrics, etc.				
Course Objective	The objective of the course is to familiarize the learners with the concepts of Advanced Natural Language Processing and attain Employability through Experiential Learning techniques.				
Course Out Comes	On successful completion of the course the students shall be able to: Understand how to solve different problems in natural language processing. [Comprehension] Solve natural language generation problems such as machine translation and text summarization. [Application] Perform sentiment analysis on reviews to discern the stance of the writer. [Application] Use public gaze behaviour data to improve the performance of different NLP systems. [Application]				
Course Content:					
Module 1	Pre-trained Language Models			4 Sessions	
Topics: Introduction to Pre-Trained Language Models. BERT. Multi-lingual variants of BERT. Introduction to NLTK and Huggingface Transformers.					
Module 2	Machine Translation and Text Summarization			7 Sessions	
Topics: Introduction to machine translation – source and target languages. Pivot-based machine translation. Using Transformers for machine translation. Monolingual machine translation examples. Machine translation evaluation metrics – BLEU. Implementation of BLEU score calculation using NLTK in Python. Other MT metrics – METEOR, TER, etc. Text summarization – definition. Types of summarizations – Extractive and Abstractive Summarization. Summarization evaluation metrics – ROUGE score.					
Module 3	Sentiment Analysis			6 Sessions	

Topics: Introduction to Sentiment Analysis. Solving sentiment analysis using text classification. Classification of sentiment analysis based on different levels – polarity-based and intensity-based. Challenges in sentiment analysis – sarcasm, thwarting, negations. Case studies in sentiment analysis – Reviewer rating prediction, short-text classifications, etc.				
Module 4	Cognitive NLP Using Gaze Behaviour			7 Sessions
Topics: Eye-Mind Hypothesis and gaze behaviour terminology. Using gaze behaviour for prediction of translation complexity, sentiment analysis complexity, sarcasm understandability, text complexity, text quality prediction, etc. Challenges with recording gaze behaviour at run time. Comparison of gaze behaviour across different people – normalization and binning. Gaze behaviour datasets. Mitigation of recording gaze behaviour at run time using type aggregation.				
List of Laboratory Tasks: Familiarization with Python. Using Python to read text files, basic tokenization and other preprocessing. Introduction to NLTK and Huggingface Transformers in Python. Using Huggingface Transformers to create a simple MT application. Implementation of pivot-based machine translation using Huggingface Transformers. Calculation of BLEU using NLTK – difference between sentence_bleu and corpus_bleu methods. Implementation of extractive summarization. Polarity classification of text using VADER. Intensity prediction of text using Weighted Normalized Polarity Intensity. Estimating gaze behaviour for a user using normalization and binning Calculating gaze behaviour for a text based on type aggregation in multiple languages. Complex word identification using gaze behaviour.				
Targeted Application & Tools that can be used: Google Colab Python IDE (Eg. PyCharm) Huggingface Transformers NLTK				
Project work/Assignment:				
Assignment: Students will have to do a course group assignment over the course of the semester. The assignment topics can be taken from Modules 2 or 3 as per the instructor-in-charge.				
Text Books				